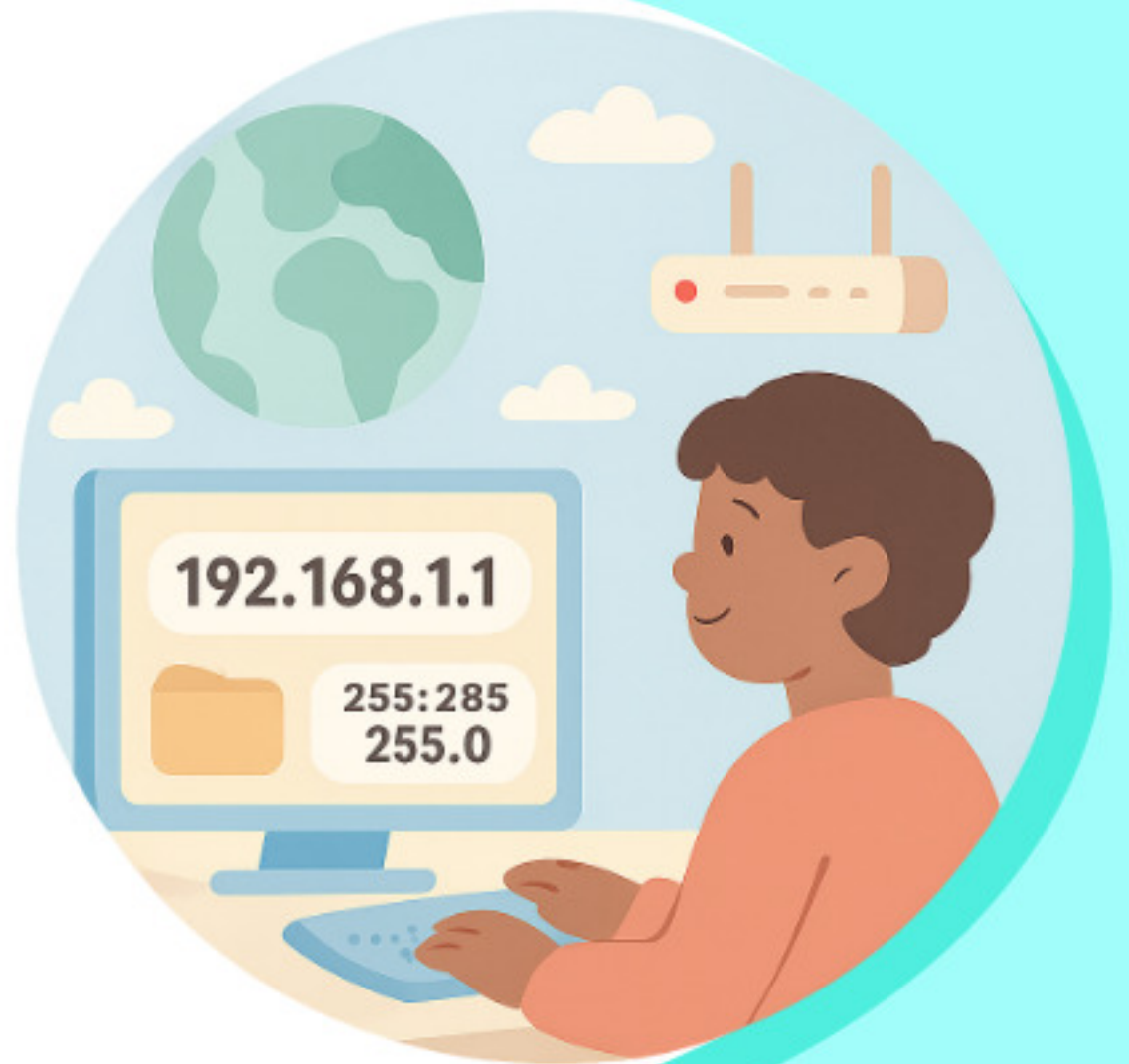


Introduction to IP Addressing



Learning Objectives

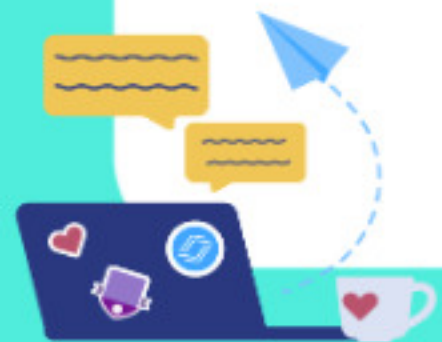
By the end of this lesson you will...

Describe the basic concepts of **IP addressing** and its significance.

Identify and explain the different **IP classes** and their uses.

Explain the role of a **default gateway** in a network and its necessity.

Describe **subnet masks** and their function in IP addressing.



Key Vocabulary

IP Address. A unique identifier assigned to each device on a network, allowing it to communicate.

Default Gateway. A device that routes traffic from a local network to other networks, enabling external communication.

Subnet Mask. A number that defines the network and host portions of an IP address, helping to organize networks.

IP Classes. Categories (A, B, C, D, E) that classify IP addresses based on their size and intended use.



Poll

How important do you think IP addresses are in networking?

- ☐ Very important, they are essential for communication.
- ☐ Somewhat important, but not the only thing.
- ☐ Not very important, I don't think about them.
- ☐ I have no idea what they are.

Introduction to IP Addressing

IP addressing is essential for identifying devices on a network, just like a **home address** helps locate your house.



What is an IP Address?

An **IP address** is a **unique identifier** for a device on a network, similar to how your **name** identifies you.



Importance of IP Addressing

IP addresses enable communication between devices over the internet, allowing us to send messages, share files, and browse websites.



IP Classes Overview

IP addresses are categorized into **classes** (A, B, C) based on size, which helps determine how many devices can connect to a **network**.



Default Gateway

A **default gateway** connects devices to **external networks**, acting like a **bridge** to the wider **internet**.



Subnet Masks

Subnet masks define the network and host portions of an IP address, helping to **organize** and **manage** networks effectively.



Understanding IP Addressing

Discuss why IP addressing is crucial in networking. Share your thoughts!

^ Instructions



Collaborate Board

Understanding IP Addressing

IP Classes Explained

IP addresses are divided into classes: **A**, **B**, **C**, **D**, and **E**. Each class serves different purposes based on **network size**.



Class A IP Addresses

Class A IP Addresses supports large networks with many hosts, making it ideal for big organizations.



Class B IP Addresses

Class B IP Addresses are designed for medium-sized networks, making them ideal for **universities** and **large businesses**. This class allows for a significant number of hosts, providing flexibility and scalability for growing organizations.



Class C IP Addresses

Class C IP addresses are designed for **small networks**, making them ideal for **small businesses** and **home networks**. This class allows for a limited number of devices, ensuring efficient communication and management.



Role of Default Gateway

The **default gateway** routes traffic from a **local network** to other networks, ensuring devices can communicate beyond their immediate area.



Why is a Default Gateway Necessary?

A **default gateway** is essential for devices to communicate beyond their local network. Without it, devices are unable to access the **internet** or other external networks, which limits their functionality and connectivity.



Open Ended Question

Explain why a default gateway is necessary in a network. Provide examples of its role.

Please enter your answer here.

What is a Subnet Mask?

A **subnet mask** determines the **network** and **host** portions of an **IP address**. It helps to define how many devices can be on a network, ensuring efficient communication and management of resources.



Function of Subnet Masks

A **subnet mask** is essential for dividing networks into smaller, manageable segments. This division enhances **organization** and **efficiency** within a network, allowing for better management of IP addresses and improved network performance.



Subnetting Example

An example of a subnet mask is **255.255.255.0**, which is commonly used for **Class C** networks. This subnet mask allows for **256 addresses** within that network, providing a range of usable IP addresses for devices.



Poll

What is the function of a subnet mask?

- ☐ To identify devices on a network
- ☐ To divide networks into segments
- ☐ To assign IP addresses
- ☐ To connect to the internet



Time To Climb

Lesson Summary

In this lesson, we reviewed several **key concepts** in networking that are essential for understanding how networks operate. These concepts include **IP addressing**, **classes**, **gateways**, and **subnet masks**.

IP addressing: A method for identifying devices on a network using unique numerical labels.

Classes: Categories of IP addresses that determine the size and type of the network.

Gateways: Devices that connect different networks and facilitate communication between them.

Subnet masks: Numbers that define the network and host portions of an IP address.

